

Ab Rauf Shah, MS, PhD., MPH-EPI ©
Lead Health Data Coordinator II | Healthcare Informatics Scientist
abrshah@unmc.edu | Omaha, Nebraska, USA
<https://scholar.google.com/citations?user=Aje0C8AAAAAJ&hl=en>

Profile Summary

Healthcare informatics and biomedical Scientist with a rich blend of experimental and digital expertise. Specializing in designing and validating liquid biopsy assays (cfDNA/ctDNA, ddPCR, NGS) and integrating multi-omics and genomic datasets to uncover actionable biomarkers in hematologic malignancies, I bring transformative insights from bench to bedside. Adept at steering cross-functional teams, I have led initiatives implementing cloud-native data platforms that seamlessly integrate EHRs, patient registries, and dynamic dashboards for real-time decision-support. My work, consistently featured in high-impact publications, bridges scientific discovery with enhanced clinical outcomes.

Key Skills

- NGS & High-Throughput Data Analysis
- Cancer Genomics & Liquid Biopsy
- Metabolic Modeling & Genomic Epidemiology
- Data Architecture & Machine Learning
- Cloud Computing (AWS) & HPC
- Cross-Functional Leadership & Collaboration
- Multicenter data governance
- Grant writing & budgets
- Mentoring & teaching

Technical Skills

- Operating Systems: Linux, macOS, Windows
- Programming Languages: Python, R, MATLAB, Java, JavaScript, Bash
- Data Analysis & Multi-Omics: RNA-seq, metabolomics, NGS variant calling, clonal evolution, metabolic model integration
- NGS & Bioinformatics Tools: GATK, STAR, Salmon, MACS2, cnvkit, DESeq2
- Statistical Packages: R (various packages), Python (NumPy, SciPy, Pandas), SPSS
- Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras, LightGBM, AutoML
- Deep Learning Applications: CNNs for image analysis, LSTMs for time-series modeling, Transformer models for genomics and NLP
- Data Science & Visualization: Jupyter, Matplotlib, Seaborn, ggplot2, Tableau, Power BI, SQL
- Wet Lab Techniques: cfDNA isolation (MagMAX, QIAamp), ddPCR (Bio-Rad QX200), qPCR, Illumina library prep, Bioanalyzer, TapeStation, cell culture
- Tools & Platforms: Git, VS Code, Docker, AWS (S3, EC2, Athena), Google Cloud, OriginLab, CI/CD

Education

- MPH-Epidemiology (in progress), University of Nebraska Medical Center, Omaha, NE, 2024–Present
- PhD., Bioinformatics, Pondicherry University, India, 2018
- MS., Bioinformatics, Jamia Millia Islamia, Delhi, India, 2010
- BSc., University of Kashmir, Srinagar, India, 2006

Certifications

- Phlebotomy Technician, Nebraska (#NE-8367-70958), 2022 – Supports liquid biopsy workflows

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Professional Experience

Lead Health Data Coordinator II, UNMC, Internal Medicine, Lymphoma Study Group (Dec 2024–Present)

- Spearhead data management for multicenter lymphoma studies with over (> 20k patients), ensuring 100% FDA compliance.
- Engineered an AWS architecture that cuts data latency by 40%, accelerating research insights and publications.
- Collaborate with clinicians, geneticists, and bioinformaticians to integrate genomic and clinical and Biomedical health-care data.
- Develop ML-powered dashboards and frameworks for EHR, digital pathology, and patient registries.

Computational Biologist, UNMC, Pathology & Microbiology (Nov 2022–Dec 2024)

- Successfully completed project Towards Digitalization of Clinical, Molecular, Digital Pathology & Cancer Genomic data. I am the developer and admin of these Apps at UNMC.
- The project is hosted at AWS & can be accessed via Combined clinical and Digital Pathology Portals:
 - <https://hemepath.unmc.edu>
 - <https://pathology.unmc.edu/apps/table.html>
- Established liquid biopsy SOPs and optimized plasma cf-DNA preservation.
- Led project to digitalize clinical, molecular, and pathology data on AWS with specialized portals.
- Developed targeted NGS panel (197 genes) for peripheral T-cell lymphoma liquid biopsy monitoring.
- Optimized ddPCR assays detecting hotspot mutations at 0.01 % VAF; validated on 120 plasma samples.
- Integrated RNA-seq, WES, and digital pathology data to identify subtype-specific vulnerabilities (ASH 2024 abstract).
- Mentored 2 graduate students in bioinformatics workflows and wet-lab validation

Research Associate | Other Hourly Worker, UNMC, Pathology & Microbiology (Feb 2022–Nov 2022)

- Conducted NGS data analysis and liquid biopsy research in cancer bioinformatics.

Volunteer Researcher, UNMC, Pathology & Microbiology (Dec 2021–Feb 2022)

- Engaged in cancer bioinformatics projects with a focus on NGS data analysis.

Postdoctoral Research Associate, University of Nebraska-Lincoln (Nov 2017–Nov 2021)

- Reconstructed and analyzed mixed-bag metabolic models of soil Viral-AMGs-bacteria-fungi.
- Developed agent-based immune system models and metabolic models of CD4+ T cells.
- Contributed to consortium grant proposals submitted to the US DOE Office of Science.

Senior Research Fellow, ICGEB, Delhi & Pondicherry University (2014–2017)

- Led marine microalgae biofuel project; scaled *Nannochloropsis* cultures to 300 L photobioreactors.
- Published first GSM of *Nannochloropsis gaditana* (iRJ1321) in Algal Research.
- Taught workshops on qPCR and metabolic modeling to postgraduate students.

Junior Research Fellow, Pondicherry University (2012–2014)

- Conducted research on culturing and characterizing marine algal strains.
- Led large-scale culturing of *Nannochloropsis* sp. for biofuel extraction.

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Research Assistant, Institute of Genomic and Integrative Biology (2010–2012)

- Enhanced research capabilities through automated literature mining.
- Created *pubmed.mineR*, R package for literature mining,
<https://www.rdocumentation.org/packages/pubmed.mineR/versions/1.0.19>
- Automated curation pipeline for PubMed abstracts improving annotation throughput by 60 %.

Teaching & Mentoring Experience

- Instructor, University of Nebraska, BIOC 435: Dynamics of Biochemical and Biological Networks (2019–2020).
- Mentored six undergraduate and graduate research projects (2015–2019) across computational systems biology and bioinformatics.

Selected Publications

- Bouska A., et al. (2025). Integrative genomic and transcriptomic analysis reveals targetable vulnerabilities in angioimmunoblastic T-cell lymphoma. *American Journal of Hematology*.
- Amador C., et al. (2025). Comprehensive Characterization of Nodal Peripheral T-Cell Lymphoma, Not Otherwise Specified. *American Journal of Clinical Pathology*.
- Hu, Z., et al. (2025). SIRT7 facilitates endometrial cancer progression by regulating PTEN stability in an estrogen-dependent manner. *Nature Communications*.
- Amador, C., et al. (2025). Refining diagnostic subtypes of peripheral T-cell lymphoma using a multiparameter approach. *Modern Pathology*.
- Lone, W., et al. (2025). High-grade B-cell lymphoma not otherwise specified, with DLBCL gene expression signatures. *American Journal of Hematology*.
- Puniya B.L., et al. Systems biology identifies drug targets in CD4⁺ T-cell immune disorders. *npj Syst. Biol. Appl.* 2021.
- **Shah AR**, et al. Genome-scale metabolic reconstruction of marine microalga *N. gaditana*. *Algal Res.* 2017.

Conferences & Invited Talks (Selection)

- ASH-2024 - Genomic Characterization of Novel PTCL-Biological Subtypes Reveal Distinctive Therapeutic Vulnerabilities
- Society for Hematopathology / EA4HP Workshop – Houston, 2023: A HemePath Portal for Clinico-Genomic Integration (Invited talk).
- ISMB 2018 – Chicago: Community Metabolic Model of Maize Root System (Oral).
- Nebraska Plant Science Retreat 2019 – Lincoln: Root-Type Specific Metabolic Models of Maize (Keynote).

Professional Memberships & Service

- American Society of Hematology (ASH) – Member
- American Society of Hematopathology – Member
- American Public Health Association – Student Member
- Manuscript reviewer for PLOS ONE, Antibiotics, Bioinformatics, npj Systems Biology
